

## Exercise 3: Complexity Theory

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**Problem 1 (Barron norm of Gaussian measure):**

Consider the density function of a Gaussian random variable,  $f(x) = (2\pi\sigma^2)^{-d/2} \exp(-\frac{\|x\|^2}{2\sigma^2})$ . We will calculate the complexity measure  $\int \|\widehat{\nabla} f(\omega)\| d\omega$  that appears in Barron's theorem discussed in class.

a) Use the following facts: (Why are they true?)

- For  $f$  defined as above, we have  $|\hat{f}(\omega)| = \exp(-2\pi^2\sigma^2\|\omega\|^2)$ ;
- (Cauchy-Schwartz inequality in expected value)  $\mathbb{E}[\|X\|] \leq (\mathbb{E}[\|X\|^2])^{\frac{1}{2}}$  where  $X$  is a random vector.

Show that  $\int \|\omega\| |\hat{f}(\omega)| d\omega \leq Z \left( \int \|\omega\|^2 Z^{-1} \hat{f}(\omega) d\omega \right)^{1/2}$  where  $Z = (2\pi\sigma^2)^{-d/2}$ .

b) Using the fact that  $\int g(\omega) Z^{-1} \hat{f}(\omega) d\omega$  is the expectation of the function  $g(\omega)$  with respect to the density  $\mathcal{N}(0, \frac{1}{4\pi^2\sigma^2})$ , show that  $\int \|\omega\| |\hat{f}(\omega)| d\omega \leq \sqrt{\frac{d}{4\pi^2\sigma^2}} \cdot (2\pi\sigma^2)^{-d/2}$ .

c) Since  $\int \|\widehat{\nabla} f(\omega)\| d\omega = 2\pi \int \|\omega\| \cdot |\hat{f}(\omega)| d\omega$ , what do you conclude about the complexity measure  $\int \|\widehat{\nabla} f(\omega)\| d\omega$  when  $d$  is very large?

**Problem 2 (Maurey's lemma):**

Let  $X = \mathbb{E}V$  be given, with  $V$  a random vector supported on a subset  $S$  of the event space, and let  $V_1, \dots, V_k$  be i.i.d. realization of  $V$ .

a) Show that

$$\mathbb{E}_{V_i} \left\| X - \frac{1}{k} \sum_{i=1}^k V_i \right\|^2 = \frac{1}{k} \mathbb{E}_V \|V - X\|^2.$$

b) Show that

$$\frac{1}{k} \mathbb{E}_V \|V - X\|^2 \leq \frac{\sup_{U \in S} \|U\|^2}{k}.$$

**Problem 3 (Number of affine pieces in a ReLU network):**

We will prove the lemma that bounds the number of affine pieces in a ReLU network stated in the lecture notes. Denote by  $\delta_A(f)$  be the (minimum) number of affine pieces of a piece-wisely linear continuous function  $f: \mathbb{R} \rightarrow \mathbb{R}$ . Show that

- a)  $\delta_A(\sum_k a_k f_k + b_k) \leq \sum_k \delta_A(f_k)$ , for any finite sequence of piece-wisely linear continuous functions  $f_k$ , and for any real sequences  $a_k$  and  $b_k$ .
- b) Let  $g_i: \mathbb{R} \rightarrow \mathbb{R}$  denote the output of some node in layer  $i$  in a ReLU network with  $L$  layers of widths  $(m_1, \dots, m_L)$  as a function of the input. Using induction on  $i$ , show that the number of affine pieces  $\delta_A(g_i)$  satisfies

$$\delta_A(g_i) \leq 2^i \prod_{j < i} m_j.$$

- c) Recall that  $g_L: \mathbb{R} \rightarrow \mathbb{R}$  is a ReLU network with  $L$  layers of widths  $(m_1, \dots, m_L)$  such that  $m = \sum_{i=1}^L m_i$ . The number of affine pieces in  $g_L$  satisfies

$$\delta_A(g_L) \leq \left( \frac{2m}{L} \right)^L.$$

- d) What value of  $L$  maximizes the upper bound  $\left( \frac{2m}{L} \right)^L$ ? Although taking this value of  $L$  gives a large value of  $\delta_A(g_L)$ , give a reason why in practice we usually pick a value much smaller than this.